

RANARISON Rina

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EDUCATION

Université d'Antananarivo, Faculté des Sciences Antananarivo, Madagascar
Master 2 — Applied Mathematics (Numerical Computation) 2025 — Present

- Relevant coursework: Error-correcting codes (BCH, Reed-Solomon, Reed-Muller, finite fields), quantum stabilizer codes (Pauli group, symplectic isomorphism), Algorithmics (P/NP/NP-completeness, SAT, graph algorithms), Numerical Methods, Applied Statistics (Data Analysis, Machine Learning)

Master 1 — Applied Mathematics (Numerical Computation) 2024 — 2025

- Relevant coursework: Partial Differential Equations, Applied Algebra (Rings, Modules), Algorithmics, Cryptography, Statistics, Computer Science, Numerical Methods

Bachelor — Mathematics (PAAC) 2021 — 2024

- Relevant coursework: Probability Theory, Statistics, Abstract Algebra, Algorithmics, Numerical Computation

Lycée Émeraude Itaosy, Antananarivo
Baccalauréat — Série C (Mathematics & Physical Sciences) 2018

EXPERIENCE

Kanty Astronomy Research Group (<https://www.zarastro.org>) Antananarivo, Madagascar
Research Assistant Internship 2026

- Student researcher, Kanty Astronomy Research Group (SAAO) — "Star Formation Activity of Arp 261"
- Analyzed HST WFPC2 imaging using ds9 and SExtractor to detect and catalogue candidate young massive star clusters
- Calibrated photometric zero-points and pixel scale from FITS headers to produce a preliminary source catalogue

Akademika (www.akademika.mg) Antananarivo, Madagascar
Co-Founder & Technical Lead 2024 — 2026

- Co-founded Madagascar's online learning platform targeting professional and career skills
- Built and deployed MVP using WordPress + TutorLMS
- Managed AWS infrastructure (EC2, S3) and a team of developers

Independent Antananarivo, Madagascar
Python Programming & Mathematics Tutor 2021 — Present

RESEARCH & INDEPENDENT PROJECTS

RSA Wiener Attack — Theory & Proof of Concept [\[PDF\]](#)
Cryptography

- Covering the mathematical theory of the Wiener attack on RSA with small private exponent; proof-of-concept implemented in Python.
- Writeup and CTF solution.

Demographic Survey Analysis of Malagasy Households [\[PDF\]](#)
Statistical Modeling

- Conducted a statistical analysis of Malagasy household demographic data using R and RStudio
- Applied descriptive statistics, visualization, and inferential methods to identify population patterns

Sentiment Analysis of Customer Feedback

[\[GitHub\]](#)

Data Science / Python

- A simple sentiment analysis to review customer feedbacks (British Airways dataset).
- Written report and Python implementation of the analysis workflow.

COMPETITIONS & DISTINCTIONS

Participant — WorldQuant IQC 2026	<i>22 Alpha Models submitted</i>	2026
Participant — Hedera Africa Hackathon	<i>Pan-African blockchain innovation</i>	2025
1st Place — BSides Madagascar CTF	<i>Cryptography & Bin. Exploitation</i>	2024
Finaliste — POESAM Orange Entrepreneurship Prize	<i>· Africa & Middle East</i>	2022
3rd Place — Weekly Coding Challenge	<i>TechZara algorithmics competition</i>	2022
3rd Place — FMSA Chess Tournament	<i>Faculty of Sciences · Université d'Antananarivo</i>	2022

TALKS & COMMUNITY

Linux Introduction — **AEMIA** 2025, 2026
Association des Étudiants en Mathématiques et Informatique · Université d'Antananarivo

- Delivered 2-hour introductory Linux sessions (theory + hands-on) to mixed-level student audiences

Pi Day & Pi Approximation Day — **AEMIA** 2024, 2025, 2026
Association des Étudiants en Mathématiques et Informatique · Université d'Antananarivo

- Served as an exhibitor at annual mathematics outreach events, presenting mathematical concepts through interactive demonstrations that made abstract ideas accessible to students and non-specialist audiences.

SKILLS

Mathematics: Abstract algebra (groups, rings, fields, modules), algebraic coding theory, numerical methods, PDEs, probability & statistics

Cryptography: RSA, AES, symmetric & asymmetric cryptography, number theory (modular arithmetic, primality)

Programming: Python (NumPy, Pandas, SciPy, PyCaret), R, C, C++, SQL, Bash

Tools & Infrastructure: AWS (EC2, S3), GNU/Linux, MAPLE, SPSS, XLStats, R Studio, Jupyter Notebook, $\text{\LaTeX}$

Languages: Malagasy (Native), French (Fluent — academic & professional), English (Advanced)

ADDITIONAL

Open Source: Added Malagasy language to MonkeyType and 10FastFingers

Hobbies: Video Games - Speed Typing · Guitar · Chess (FIDE elo 1754)